Traverse Closure Report: Example site control loop

Project	Example site control loop	Date	2026-10-23
Party chief	-	Software	Traverse Doctor v0.1.0
Start point	A: N 5,000.000, E 5,000.000	Start direction	S 49°19'13" E (A-B)

MEETS REQUIREMENTS. Precision 1:110,251 (required 1:10,000). No blunder found. The traverse meets the required precision.

Summary

The 5-station loop has a perimeter of 3329.36 ft. The angular misclosure is +2.0" against an allowable of 34", so each angle was corrected by -0.4". After balancing, the linear misclosure is 0.03 ft toward S 62°43'08" W, a precision of 1:110,251 against the required 1:10,000.

The least squares adjustment has 3 redundant observations and a reference variance of 0.431, which passes the chi-square test. Coordinates in this report come from the least squares adjustment, and the area is 756,971.9 sq ft (17.3777 ac).

The traverse meets the requirements and no blunder was found.

Template summary (no Claude API key set).

Angular closure

Angles sum to 540°00'02". Interior angles of a 5-sided loop should sum to 540°00'00". Misclosure +2.0", allowable 3 x 5" x sqrt(5) = 34". Correction per angle -0.40".

Station	Angle right	Balanced	Course	Azimuth	Bearing	Distance (ft)
A	106°15'57"	106°15'56.6"	A-B	130°40'47.0"	S 49°19'13.0" E	641.400
B	113°23'03"	113°23'02.6"	B-C	64°03'49.6"	N 64°03'49.6" E	705.160
C	96°45'44"	96°45'43.6"	C-D	340°49'33.2"	N 19°10'26.8" W	658.260
D	120°19'45"	120°19'44.6"	D-E	281°09'17.8"	N 78°50'42.2" W	628.610
E	103°15'33"	103°15'32.6"	E-A	204°24'50.4"	S 24°24'50.4" W	695.930

Latitudes, departures and compass rule

Course	Latitude	Departure	Corr. lat	Corr. dep	Adj. distance	Adj. bearing
A-B	-418.084	486.415	+0.0027	+0.0052	641.402	S 49°19'15" E
B-C	308.416	634.137	+0.0029	+0.0057	705.166	N 64°03'50" E
C-D	621.743	-216.199	+0.0027	+0.0053	658.261	N 19°10'25" W
D-E	121.613	-616.734	+0.0026	+0.0051	628.606	N 78°50'41" W
E-A	-633.702	-287.647	+0.0029	+0.0056	695.925	S 24°24'49" W
Sum	-0.0138	-0.0268	+0.0138	+0.0268	3,329.360	

Linear misclosure = sqrt(-0.0138² + -0.0268²) = 0.0302 ft toward S 62°43'08" W. Precision = 3,329.360 / 0.0302 = 1:110,251.

Adjusted coordinates

Station	Compass N	Compass E	Transit N	Transit E	LSQ N	LSQ E
A	5,000.000	5,000.000	5,000.000	5,000.000	5,000.000	5,000.000
B	4,581.919	5,486.421	4,581.919	5,486.421	4,581.915	5,486.417

Station	Compass N	Compass E	Transit N	Transit E	LSQ N	LSQ E
C	4,890.338	6,120.564	4,890.337	6,120.566	4,890.326	6,120.561
D	5,512.084	5,904.370	5,512.084	5,904.370	5,512.075	5,904.376
E	5,633.699	5,287.641	5,633.698	5,287.643	5,633.698	5,287.648

Area by coordinates (Least squares): 756,971.9 sq ft (17.3777 ac). Compass 756,972.0, transit 756,972.3, least squares 756,971.9 sq ft.

Least squares adjustment

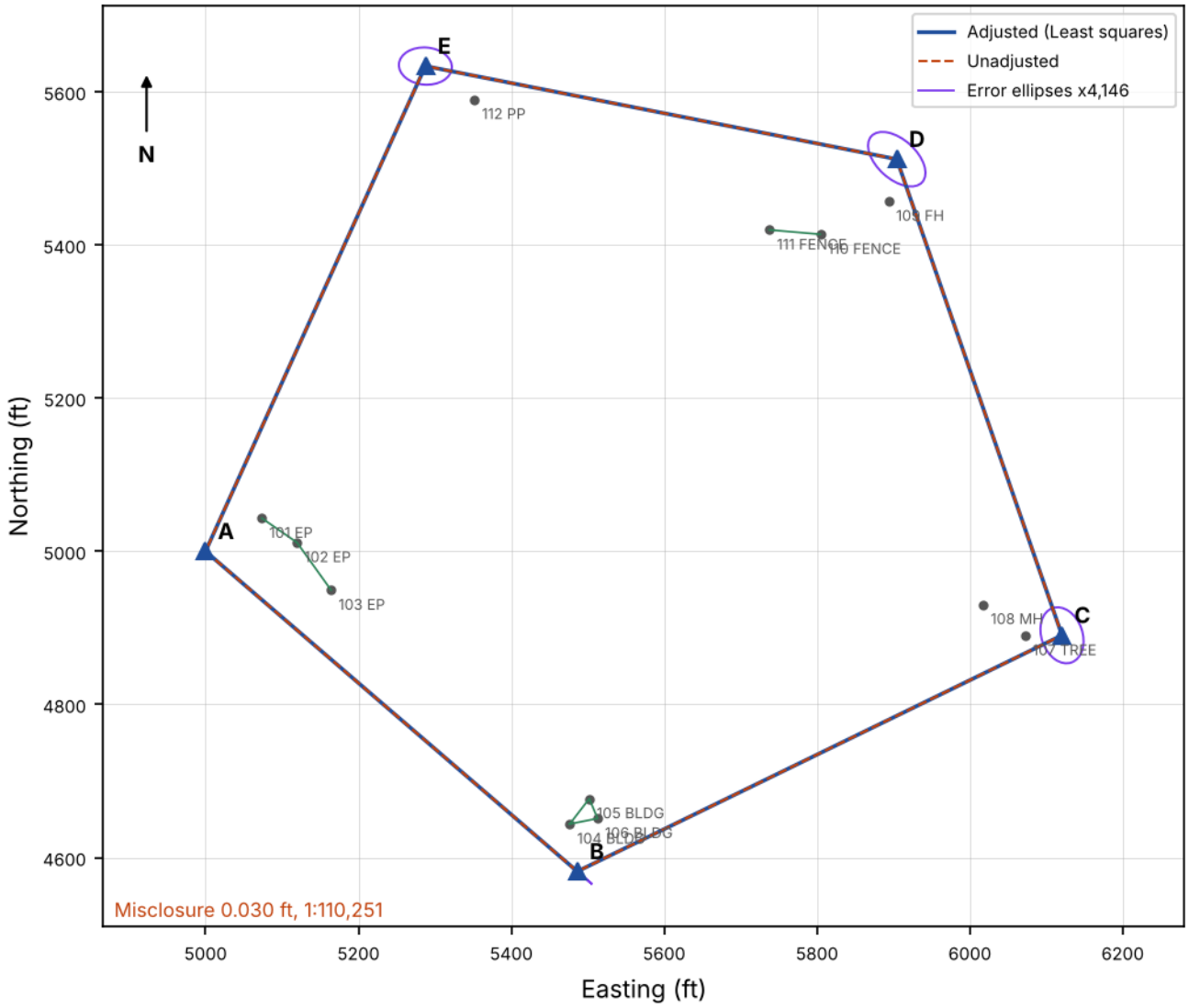
Weights from sigma angle 5" and sigma distance 0.01 ft + 3 ppm. A held fixed with the direction of A-B. Converged in 2 iterations. Redundancy 3, reference variance 0.431 (sigma0 0.657). Chi-square test at 95%: 0.072 to 3.116, passes. Standardized residuals over 3.29 are flagged.

Observation	Observed	Residual	Std. residual	Redundancy	Flag
Angle at A	106°15'57"	-1.2"	-0.34	0.46	
Angle at B	113°23'03"	+1.9"	+0.56	0.44	
Angle at C	96°45'44"	+2.0"	+0.55	0.50	
Angle at D	120°19'45"	-1.3"	-0.42	0.42	
Angle at E	103°15'33"	-3.3"	-0.97	0.47	
Distance A-B	641.400	+0.0016	+0.42	0.14	
Distance B-C	705.160	+0.0044	+1.12	0.14	
Distance C-D	658.260	+0.0006	+0.16	0.14	
Distance D-E	628.610	-0.0034	-0.88	0.14	
Distance E-A	695.930	-0.0034	-0.89	0.14	

Standard error ellipses (scaled by the reference variance)

Station	Sigma N (ft)	Sigma E (ft)	Semi-major	Semi-minor	Major axis az.
A	0.0000	0.0000	0.0000	0.0000	fixed
B	0.0040	0.0047	0.0062	0.0000	130°40'47"
C	0.0088	0.0068	0.0091	0.0065	160°45'53"
D	0.0086	0.0090	0.0108	0.0062	132°01'38"
E	0.0059	0.0084	0.0084	0.0059	94°21'26"

Plan



The loop runs counterclockwise on the plot. If it looks mirrored compared with your sketch, the angles were turned the other way: switch between angles right and angles left.

Point list (from the least squares coordinates)

Point	Northing	Easting	Code	Layer	Description
A	5,000.000	5,000.000	CP	V-CTRL	Mag nail in curb
B	4,581.915	5,486.417	IPS	V-PROP-MON	5/8 in rebar set
C	4,890.326	6,120.561	IPS	V-PROP-MON	5/8 in rebar set
D	5,512.075	5,904.376	CP	V-CTRL	PK nail in asphalt
E	5,633.698	5,287.648	IPF	V-PROP-MON	1/2 in pipe found
101	5,043.243	5,073.666	EP1 B	V-ROAD-EDGE	Edge of pavement
102	5,011.518	5,119.627	EP1	V-ROAD-EDGE	
103	4,949.318	5,164.259	EP1 E	V-ROAD-EDGE	
104	4,643.873	5,476.479	BLDG B	V-BLDG	Building corner
105	4,676.936	5,502.061	BLDG	V-BLDG	

Point	Northing	Easting	Code	Layer	Description
106	4,651.139	5,512.909	BLDG C	V-BLDG	
107	4,889.960	6,072.363	TREE	V-SITE-VEG	24 in oak
108	4,930.612	6,017.613	MH	V-SSWR-MH	Sanitary manhole
109	5,457.275	5,893.858	HYD	V-WATR-HYDR	Fire hydrant
110	5,414.026	5,804.303	FENCE B	V-SITE-FENC	Chain link
111	5,419.743	5,737.839	FENCE	V-SITE-FENC	
112	5,590.182	5,351.959	PP	V-POWR-POLE	Power pole

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